Tunneling Through the Fog for Long Distance Visibility

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Objective

To create a "tunnel" through fog by heating suspended droplets using an appropriate radiation energy spectrum.

Figure: Fog causing traffic delays

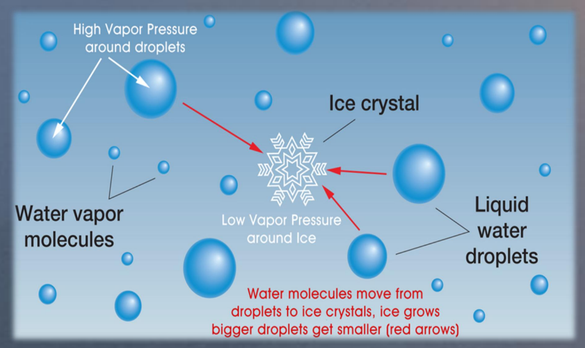

Figure: constituents of Fog

Optical Properties of Droplet/Ice

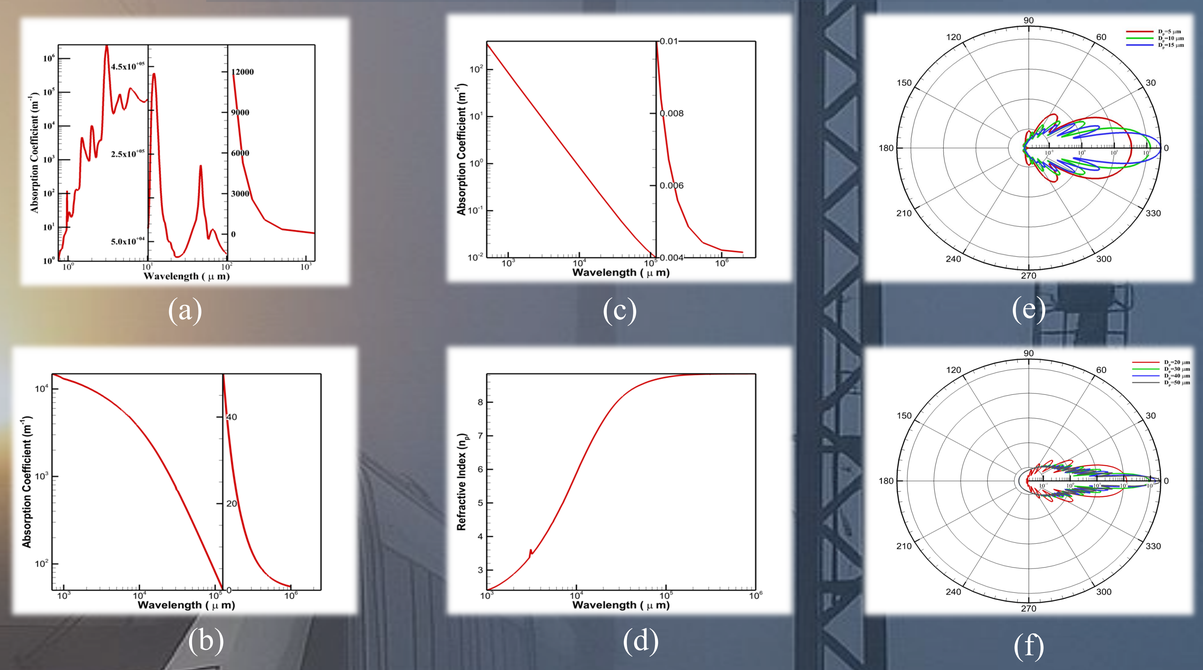

Figure: Variation of the absorption coefficient and scattering phase function of water and ice in the visible, infrared and microwave spectra

Computational Modelling

- **Continuity Equation:**

$$\frac{\partial U_i}{\partial x_i} = -\frac{1}{\rho}\frac{d\rho}{dt}$$

- **Volume Fraction Equation:**

$$\frac{\partial \alpha_q}{\partial t} + \frac{\partial}{\partial x_j}\left(\alpha_q u_j\right) = \frac{\dot{m}_q''}{\rho_q}$$

$$\sum_q \alpha_q = 1$$

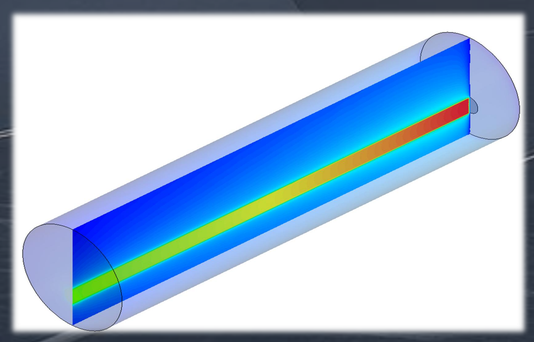

Figure: geometry for simulation

- **Momentum Equation:**

$$\frac{\partial(\rho u_i)}{\partial t} + \frac{\partial}{\partial x_j}\left(\rho u_i u_j\right) = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j}\left[\mu\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)\right] + \rho g_i + (F_s)_i$$

- **Energy Equation:**

$$\frac{\partial(\rho h)}{\partial t} + \frac{\partial(\rho h u_j)}{\partial x_j} = \lambda\frac{\partial^2 T}{\partial x_j \partial x_j} + q'' + S_R$$

$$\rho = \sum_q \alpha_q \rho_q, \quad \mu = \sum_q \alpha_q \mu_q, \lambda = \sum_q \lambda_q \mu_q$$

- **Radiation Modelling:**

Monte Carlo method (Statistical Method)

Contours

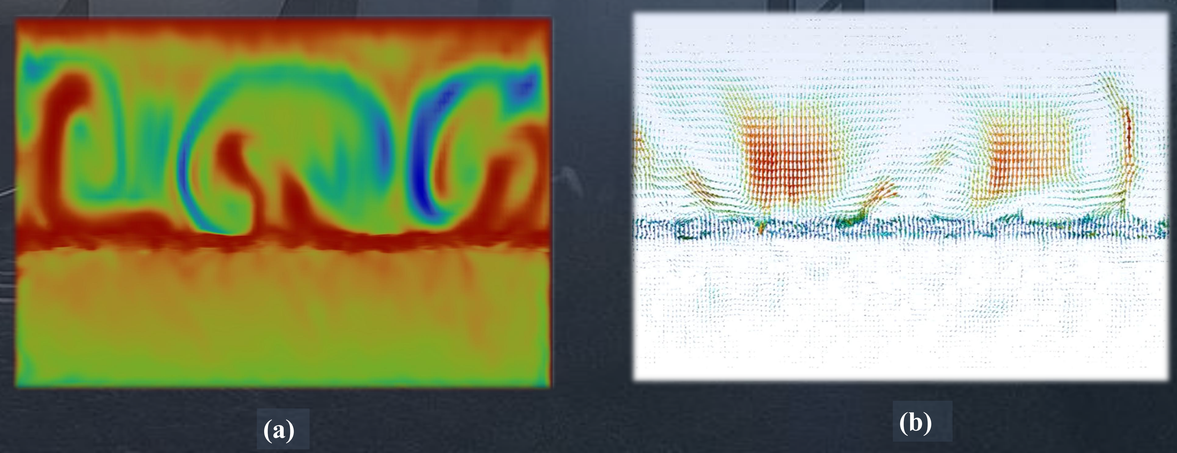

Figure: Contour a) Volume Fraction and b) Velocity Vectors

Conclusion

- With the appropriate selection of spectrum for the radiation energy, creating a tunnel in fog is possible.

References

- Warren, S. G., and R. E. Brandt (2008), Optical constants of ice from the ultraviolet to the microwave: A revised compilation, J. Geophys. Res., 113, D14220, doi:10.1029/2007JD009744.
- Seglestein D.J. The complex refractive index of water 1981; M.S. dissertation: 167.